

Expt 6	Finite Word length Effects
Date:	6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

CODE :-

```

clc;

clear;

close all;

%% Signal

fs = 1000;

f = 50;

t = 0:1/fs:0.1;

x = sin(2*pi*f*t);

%% Quantization Levels

L = [8 16 32];

colors = {'r','g','m'};

figure;

for i = 1:length(L)

    % Quantization (FIXED LINE)

    xq{i} = round((x + 1) * (L(i) - 1) / 2) * (2 / (L(i) - 1)) - 1;

    % Error & MSE

    e{i} = x - xq{i};

```

```
mse(i) = mean(e{i}.^2);
```

```
% Plot Quantized Signals
```

```
subplot(4,1,i+1);
```

```
stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);
```

```
title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
```

```
grid on;
```

```
end
```

```
%% Original Signal
```

```
subplot(4,1,1);
```

```
plot(t, x, 'LineWidth', 1.5);
```

```
title('Original Signal');
```

```
grid on;
```

```
%% Display MSE
```

```
fprintf('MSE for 8 Levels = %f\n', mse(1));
```

```
fprintf('MSE for 16 Levels = %f\n', mse(2));
```

```
fprintf('MSE for 32 Levels = %f\n', mse(3));
```

```
%% Error Plots
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i);
```

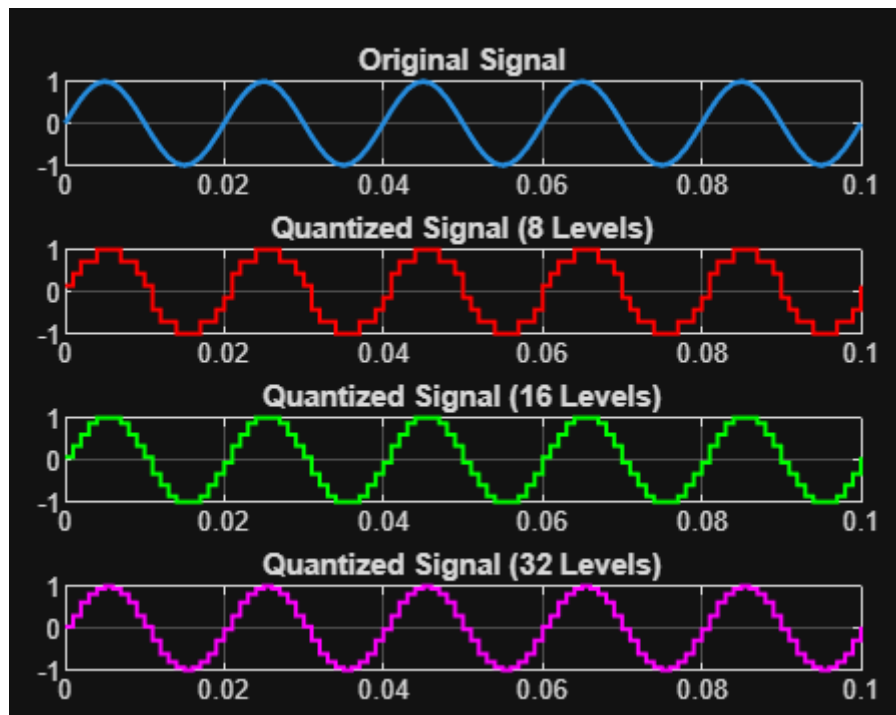
```
    plot(t, e{i});
```

```
    title(['Error (' num2str(L(i)) ' Levels)']);
```

```
    grid on;
```

```
end
```

OUTPUT :-



MSE for 8 Levels = 0.010473

MSE for 16 Levels = 0.001763

MSE for 32 Levels = 0.000357

